

of 20 mm, and fill it with the colour red. The important thing to note here is that vectors rely on real-world dimensions, while raster images are all pixel-based. Thus Windows XP would draw a line 50 pixels long, while Windows Vista would draw a line 5 cm long—the difference here is that when you increase the resolution of your monitor, the 50 pixels line would appear smaller because the pixels are now much closer to each other, but the 5 cm line would still be 5 cm long. Thus a vector-based interface can be resolution and pixel-density independent.

Secondly, Vista offers hardware-accelerated support for this API: through the 3D interfaces of DirectX 9 and 10, Vista can leverage modern video cards to simultaneously offer a visually rich and fast interface.

Along with the graphics API, Vista also throws out the old window manager and the windowing API. This total overhaul is termed the Windows Presentation Foundation.

As part of this package, Vista introduces a new window manager called the Desktop Window Manager (DWM). The DWM draws each window to its own buffer, thus tracking what each application is displaying at all times. These individual buffers can be filled, coloured, transformed, distorted, overlaid, and so on, to create all manner of effects

Windows Presentation Foundation offer support to 2D and 3D shapes. It is also vector-based and hardware accelerated. WPF also removes absolute units for measuring onscreen elements: while Windows XP uses pixels to draw up screen elements, Vista goes for real-world units—thus, a button would be one-inch tall, or a line 4 cm long, or a font 8 points big, and so on. This ensures that the size of the on-screen element remains the same even if the pixel density of the screen increases or decreases.

One of the issues that programmers and designers of applications face today is that there is no clear way to separate the